

Single Agent Systems & Agent Pipelines

- Quiz

Instructions:

Answer the following questions. These questions are designed to test your understanding of agent systems, tools, and evaluation.

1. Explain the concept of a stateful directed graph in agent pipelines. How does it differ from a simple linear pipeline?

Stateful directed graph refers to a type of workflow in which all of the steps are able to save and make use of data from previous steps. The workflow includes nodes and edges which describe the flow of data within the workflow. Branching, loops, and decision-making capabilities make it smarter and more flexible. On the other hand, linear pipeline does not have the ability to change or remember previous states while performing a task.

2. Describe the role of nodes and edges in an agent workflow. Give an example of each.

The nodes are the steps or the activities carried out by an agent. The nodes could be processing of a request, invocation of a tool, or sending of a response among others. The edges on the other hand are connections from one node to another and determine the flow of information. For instance, Calculator Tool could be a node while the route to the calculator from query analysis would be the edge.

3. What is conditional routing in an agent system? Design a simple rule-based routing logic for three different query types.

Conditional routing involves the redirection of a query to various tools depending on its intention. This enables the agent to select the best approach for generating responses to the query. In case the query includes the term "calculate," it will be redirected to the Calculator Tool; whereas the inclusion of the term "keywords" will lead to redirecting to the Keyword Extraction Tool.

4. Why are cycles (loops) important in agent pipelines? Provide a use case where a retry loop is necessary.

Cycles or Loops help an agent iterate through a procedure until a certain condition is met. Loops come in handy where actions may fail or where an action has to be done several times for success. In case of an API call that does not succeed because of some temporary connectivity problem, the agent can attempt the call again rather than failing right away.

5. Explain how a single-agent system can simulate multi-agent behavior internally.

The tasks of many agents can be carried out by a single-agent system, and they can be done by breaking down the task into different functions. The system can begin by analyzing the query, determine the right

tool to use, and finally generate an answer. Even though all of the above functions are performed by a single system, each of them works independently like an agent.

6. What are JSON schema tools? How do they help in structuring tool inputs and outputs?

The JSON schema tool provides an established structure for the data passed between the agent and tools. The tool provides information on what is necessary and expected from the input data. This means that before any processes occur, the inputs should be appropriate. This helps to ensure the consistency of output data.

7. Compare sequential tool calls and parallel tool calls. When would you prefer one over the other?

Sequence tool call involves one task following another in such a way that every task is dependent on the output of the previous one. Parallel tool call involves carrying out several independent processes concurrently. Sequence calls are appropriate when there are task dependencies. Parallel calls are best suited for tasks that are independent and can run at once.

8. How would you implement error handling in a tool-using agent? Provide at least two strategies.

Error handling enables an agent to keep on running despite any problems. One approach is the use of try-except constructs to catch any exceptions and provide relevant error messages. Other approaches include the implementation of retry functions that will allow for automatic rerunning of any actions that fail. This allows for effective debugging and logging of errors.

9. What is trajectory evaluation in agent systems? Why is it important beyond just checking final output?

The concept of trajectory evaluation involves the study of the whole path of action taken by an agent while solving a particular task. It considers decision-making, the call made for the use of a tool, and other intermediary actions that an agent performs during the process of solving a task. Trajectory evaluation allows us to discover errors that would otherwise be hidden in the final solution of a task.

10. Define task completion rate and cost metrics. How would you measure and optimize them in a real-world system?

The task completion rate indicates how many tasks are successfully accomplished by the agent. Cost measures the resource utilization, which could be the number of API calls or processing time spent on the process. This metric could be determined by monitoring agent performance while accomplishing many tasks. They could be improved by improving routing efficiency, minimizing tool calls, and employing more efficient algorithms.